Quantum for Optimisation

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Structure

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\begin{aligned}
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  so that $(f(z_i))_{i=\dots}$ are **eigenvalues** of $\mathcal{H}_F$.
- **Solving the minimisation problem** is thus equivalent to finding the smallest eigenvalues (minimum energy) of $\mathcal{H}_F$.
- **Problem:** it is often difficult to find them.

Variational Quantum Eigensolver

Variational Quantum Eigensolver

The **Variational Quantum Eigensolver** (VQE) is a PQC-based algorithm that aims to find the smallest eigenvalue (the lowest energy) of a problem Hamiltonian.

Objective functions of many NP-hard combinatorial optimisation problems can be encoded in the Hamiltonians of the quantum systems – finding the **ground state** of the Hamiltonian gives us the minimum of the objective function.

The **variational** part of the algorithm refers to the systematic search for the best possible approximation of the ground state by trying various PQC ansatzes and configurations of adjustable PQC parameters – the variational approach.

How does the algorithm work?

The *characteristic equation* for the Hamiltonian $\mathcal{H}$ is

$$\mathcal{H}|\psi_i\rangle = \lambda_i|\psi_i\rangle,$$

where $|\psi_i\rangle$ is an eigenstate associated with the eigenvalue λ_i.

The objective is to find the smallest eigenvalue λ_0 (the lowest energy) of $\mathcal{H}$ corresponding to the ground state (the lowest energy state) $|\psi_0\rangle$.

The eigenvalue (energy) of $\mathcal{H}$ is the expectation of $\mathcal{H}$:

$$\langle\psi_i|\mathcal{H}|\psi_i\rangle = \langle\psi_i| \cdot \lambda_i \cdot |\psi\rangle_i = \lambda_i \langle\psi_i|\psi_i\rangle = \lambda_i.$$

This expectation can be calculated on a quantum computer.

Therefore, the task is to use the PQC to construct a quantum state that is as close as possible to the ground state of the problem Hamiltonian.

We do it by trying many candidate states and choosing the one that minimises the expectation value.

Motivation: Spectral theorem

Since a Hamiltonian $\mathcal{H}$ is Hermitian, it can be expressed in a basis $\{b_1, b_2, \dots\}$:

$$\mathcal{H} = \sum_i \lambda_i |b_i\rangle \langle b_i| = \sum_i \lambda_i \mathcal{H}_i, \qquad \text{with } 0 \leq \lambda_0 \leq \lambda_1 \leq \cdots$$

and $\mathcal{H}_i = |b_i\rangle \langle b_i|$ the projection onto the eigenspace of $\mathcal{H}$ corresponding to λ_i.

Born's rule

When measuring an observable $\mathcal{H}$ in the system $|\psi\rangle$, the probability of obtaining a given eigenvalue λ_i is equal to $\langle\psi|\mathcal{H}_i|\psi\rangle$.

Indeed, expanding $|\psi\rangle = \sum_k \alpha_k |b_k\rangle$, so that $\langle\psi| = \sum_j \alpha_j^* \langle b_j|$,

$$\langle\psi|\mathcal{H}_i|\psi\rangle = \left\langle \sum_j \alpha_j b_j \middle| \mathcal{H}_i \middle| \sum_k \alpha_k |b_k\rangle \right\rangle = \sum_{j,k} \alpha_j^* \alpha_k \langle b_j|\mathcal{H}_i|b_k\rangle = \sum_j \alpha_j^* \alpha_i \langle b_j|b_i\rangle = |\alpha_i|^2.$$

We can then compute

$$\langle\psi|\mathcal{H}|\psi\rangle = \left\langle \psi \middle| \sum_i \lambda_i \mathcal{H}_i \middle| \psi \right\rangle = \sum_i \lambda_i \left\langle \psi \middle| \mathcal{H}_i \middle| \psi \right\rangle = \sum_i \lambda_i |\alpha_i|^2 \geq \lambda_0 \sum_i |\alpha_i|^2 = \lambda_0.$$

The hybrid quantum-classical protocol

- **Role of the PQC**: produce the candidate states $|\psi\rangle$.
- **The variational part** of the algorithm consists of iterative improvements of the candidate state (iterative updates of the adjustable parameters).
- **The quantum part** of the algorithm consists of running the PQC and then measuring $\mathcal{H}$ on the constructed quantum state to obtain the expectation of $\mathcal{H}$.

Computing quantum expectations (one qubit)

Any 2×2 (one-qubit) unitary and Hermitian matrix can be decomposed into a sum of Pauli matrices $\mathtt{X}, \mathtt{Y}, \mathtt{Z}, \mathtt{I}$:

$$\mathcal{H} = a\mathtt{X} + b\mathtt{Y} + c\mathtt{Z} + d\mathtt{I},$$

for $a, b, c, d \in \mathbb{R}$. For a given $|\psi\rangle$, the expectation of a Hamiltonian $\mathcal{H}$ reads

$$\langle\mathcal{H}\rangle := \langle\psi|\mathcal{H}|\psi\rangle = a \langle\psi| \mathtt{X} |\psi\rangle + b \langle\psi| \mathtt{Y} |\psi\rangle + c \langle\psi| \mathtt{Z} |\psi\rangle + d \langle\psi| \mathtt{I} |\psi\rangle .$$

Thus we can compute the expectation values of the Pauli terms independently and then sum them up to obtain $\langle\mathcal{H}\rangle$.

Note: Again, the computations will be up to (Monte Carlo) statistical errors.

Expectation values of Pauli operators

Expectation value of $\mathtt{I}$ is 1: $\langle\psi|\,\mathtt{I}\,|\psi\rangle = \langle\psi|\psi\rangle = 1$.
$\rightarrow$ *This term will contribute d to $\langle\mathcal{H}\rangle$.*

Next term: $c\mathtt{Z}$. Write

$$|\psi\rangle = \alpha_z\,|0\rangle + \beta_z\,|1\rangle\,,$$

for $\alpha_z, \beta_z \in \mathbb{C} : |\alpha_z|^2 + |\beta_z|^2 = 1$. Then (recall that $\mathtt{Z}\,|1\rangle = -\,|1\rangle$, $\mathtt{Z}\,|0\rangle = |0\rangle$)

$$\begin{aligned}\langle\psi|\mathtt{Z}|\psi\rangle &= \Big(\alpha_z^*\,\langle 0| + \beta_z^*\,\langle 1|\Big)\,\mathtt{Z}\,\Big(\alpha_z\,|0\rangle + \beta_z\,|1\rangle\Big)\\ &= |\alpha_z|^2\,\langle 0|\,\mathtt{Z}\,|0\rangle \quad +\alpha_z^*\beta_z\,\langle 0|\,\mathtt{Z}\,|1\rangle \quad +\alpha_z\beta_z^*\,\langle 1|\,\mathtt{Z}\,|0\rangle \quad +|\beta_z|^2\,\langle 1|\,\mathtt{Z}\,|1\rangle\\ &= |\alpha_z|^2\,\langle 0|0\rangle \quad -\alpha_z^*\beta_z\,\langle 01\rangle \quad +\alpha_z\beta_z^*\,\langle 10\rangle \quad -|\beta_z|^2\,\langle 11\rangle\\ &= |\alpha_z|^2 \quad +0 \quad +0 \quad -|\beta_z|^2,\end{aligned}$$

where $|\alpha_z|^2$ and $|\beta_z|^2$ are the probabilities that after z-basis measurement the quantum state $|\psi\rangle$ will become $|0\rangle$ or $|1\rangle$ respectively.

Expectation value of Z

- Run the Quantum circuit to construct $|\psi\rangle$ and perform N measurements;
- Probability of finding qubit in "0" estimated by $\frac{n_0}{N}$, with n_0 the number of "0" observations.
- Probability of finding qubit in state "1" $\approx \frac{n_1}{N}$;
- $n_0 + n_1 = N$;
- Contribution of the Z-term to $\langle \mathcal{H} \rangle$ is given by

$$c \langle\psi|\, \mathrm{Z}\, |\psi\rangle = c\frac{n_0 - n_1}{N}.$$

Expectation values of $\mathtt{X}$ and $\mathtt{Y}$

States $|0\rangle$ and $|1\rangle$ are the eigenstates of $\mathtt{Z}$ with eigenvalues $+1$ and -1:

$$\mathtt{Z}\,|0\rangle = |0\rangle\,, \quad \mathtt{Z}\,|1\rangle = -\,|1\rangle\,.$$

The eigenstates of operator $\mathtt{X}$ are

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{and} \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

and the eigenstates of operator $\mathtt{Y}$ are

$$|\mathrm{R}\rangle = \frac{|0\rangle + \mathrm{i}\,|1\rangle}{\sqrt{2}} \quad \text{and} \quad |\mathrm{L}\rangle = \frac{|0\rangle - \mathrm{i}\,|1\rangle}{\sqrt{2}}.$$

Their corresponding eigenvalues also are $+1$ and -1:

$$\mathtt{X}|+\rangle = |+\rangle, \quad \mathtt{X}|-\rangle = -|-\rangle, \qquad \mathtt{Y}|\mathrm{R}\rangle = |\mathrm{R}\rangle, \quad \mathtt{Y}|\mathrm{L}\rangle = -|\mathrm{L}\rangle.$$

Expectation values of $\mathtt{X}$ and $\mathtt{Y}$ (continued)

Therefore, the quantum state $|\psi\rangle$ can also be decomposed into the superposition of the basis states $\{|\mathrm{R}\rangle, |\mathrm{L}\rangle\}$ (y-basis) and $\{|+\rangle, |-\rangle\}$ (x-basis):

$$|\psi\rangle = \alpha_x|+\rangle + \beta_x|-\rangle = \alpha_y|\mathrm{R}\rangle + \beta_y|\mathrm{L}\rangle.$$

If we can perform measurement in x-basis and y-basis, the expectations $\langle\psi|\,\mathtt{X}\,|\psi\rangle$ and $\langle\psi|\,\mathtt{Y}\,|\psi\rangle$ can be calculated in exactly the same way as expectation $\langle\psi|\,\mathtt{Z}\,|\psi\rangle$:

$$a\,\langle\psi|\,\mathtt{X}\,|\psi\rangle = a\frac{n_+ - n_-}{N}, \quad b\,\langle\psi|\,\mathtt{Y}\,|\psi\rangle = b\frac{n_\mathrm{R} - n_\mathrm{L}}{N}.$$

- n_+ and n_- are the numbers of measurements in the x-basis corresponding to the $|+\rangle$ and $|-\rangle$ outcomes.
- n_R and n_L are the numbers of measurements in the y-basis corresponding to the $|\mathrm{R}\rangle$ and $|\mathrm{L}\rangle$ outcomes.

Expectation values of X and Y (continued)

If we can only perform measurement in the z-basis, we need to apply some additional operators (gates) to state $|\psi\rangle$ before the measurement, such that the probability of measuring $|0\rangle$ in z-basis is the same as the probability of measuring $|+\rangle$ in x-basis if we are calculating $\langle\psi|\,\mathtt{X}\,|\psi\rangle$, or the probability of measuring $|0\rangle$ in z-basis is the same as the probability of measuring $|\mathrm{R}\rangle$ in y-basis if we are calculating $\langle\psi|\,\mathtt{Y}\,|\psi\rangle$:

$$\mathtt{H}\,|\psi\rangle = \mathtt{H}\,(\alpha_x|+\rangle + \beta_x|-\rangle) = \alpha_x\,|0\rangle + \beta_x\,|1\rangle\,,$$

with $\mathtt{H}|+\rangle = |0\rangle$ and $\mathtt{H}|-\rangle = |1\rangle$.

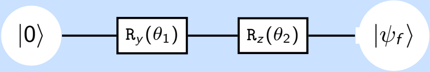
$$\mathtt{G}\,|\psi\rangle = \mathtt{G}\,(\alpha_y|\mathrm{R}\rangle + \beta_y|\mathrm{L}\rangle) = \alpha_y\,|0\rangle + \beta_y\,|1\rangle\,,$$

with $\mathtt{G}|\mathrm{R}\rangle = |0\rangle$ and $\mathtt{G}|\mathrm{L}\rangle = |1\rangle$.

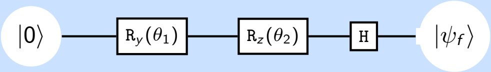
The operators H and G are the following one-qubit rotations:

$$\mathtt{H} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \mathtt{G} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -\mathrm{i} \\ 1 & \mathrm{i} \end{bmatrix}.$$

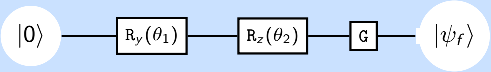
Constructing the PQC (continued)



PQC for a one-qubit system to calculate $\langle Z \rangle$.



PQC with H gate to calculate $\langle X \rangle$.



PQC with G gate to calculate $\langle Y \rangle$.

Quantum Approximate Optimisation Algorithm

The time evolution

The dynamics of the quantum system is governed by the Schrödinger equation

$$\mathrm{i}\hbar\frac{\mathrm{d}|\psi(t)\rangle}{\mathrm{d}t} = \mathcal{H}|\psi(t)\rangle,$$

where $|\psi(t)\rangle$ is the quantum state at time t and $\mathcal{H}$ is the *time-independent* Hamiltonian. The solution of Schrödinger equation is given by

$$|\psi(t)\rangle = \mathcal{U}(0,t)|\psi(0)\rangle,$$

where the operator $\mathcal{U}(t)$ is obtained from the Hamiltonian $\mathcal{H}$ via

$$\mathcal{U}(0,t) = \exp\left(\frac{-\mathrm{i}\mathcal{H}t}{\hbar}\right).$$

We normalise to $\hbar = 1$, so that

$$|\psi(t)\rangle = \mathrm{e}^{-\mathrm{i}\mathcal{H}t}|\psi(0)\rangle.$$

Time-dependent Hamiltonian

If the initial state $|\psi(0)\rangle$ is known then the state of the system at time t is also known and is determined by the action of Hamiltonian $\mathcal{H}$ over a period of length t.

Ideally, we would like to work with the time-dependent Hamiltonians of the form

$$\mathcal{H}(t) = \left(1 - \frac{t}{T}\right) \mathcal{H}_0 + \frac{t}{T} \mathcal{H}_F,$$

where $\mathcal{H}_0$ is the *initial* Hamiltonian and $\mathcal{H}_F$ is the *final* or *problem* (i.e., encoding the optimisation problem) Hamiltonian.

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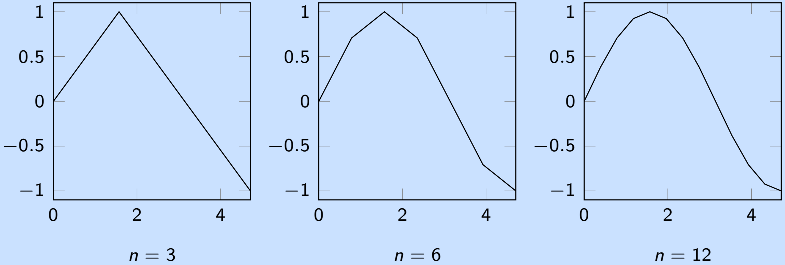
We can *approximate* $\mathcal{H}(t)$ by a sequence of time-independent Hamiltonians $\{\mathcal{H}_1, \mathcal{H}_2, \ldots, \mathcal{H}_n\}$, transforming the state in corresponding shorter time intervals $\{[t_0 = 0, t_1], [t_1, t_2], \ldots, [t_{n-1}, t_n = T]\}$.

Approximation

Analogy: approximation of continuous function by a piecewise linear function. The more granular the time intervals $[t_{i-1}, t_i]$, the better the approximation.



Piecewise linear approximation of $\sin(t)$.

Similarly, we can approximate operator $\mathcal{U}(0, T)$ as

$$\mathcal{U}(0, T) = \mathcal{U}(t_{n-1}, t_n = T)\mathcal{U}(t_{n-2}, t_{n-1}) \cdots \mathcal{U}(t_2, t_1)\mathcal{U}(t_0 = 0, t_1).$$

The Suzuki-Trotter expansion

A useful approximation of $\mathcal{U}(0, T)$ can be obtained using the *Suzuki-Trotter expansion*. If $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_p$ are operators that do not necessarily *commute*, then

$$\mathrm{e}^{\mathcal{A}_1+\mathcal{A}_2+\ldots+\mathcal{A}_p} = \lim_{m\uparrow\infty}\left(\mathrm{e}^{\frac{\mathcal{A}_1}{m}}\mathrm{e}^{\frac{\mathcal{A}_2}{m}}\cdots\mathrm{e}^{\frac{\mathcal{A}_p}{m}}\right)^m$$

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Two operators $\mathcal{A}$ and $\mathcal{B}$ are said to commute with each other if

$$\mathcal{A}\mathcal{B} - \mathcal{B}\mathcal{A} = 0,$$

or equivalently $\mathcal{A}\mathcal{B}\left|\psi\right\rangle = \mathcal{B}\mathcal{A}\left|\psi\right\rangle$ for any quantum state $\left|\psi\right\rangle$.

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or equivalently $\mathcal{A}\mathcal{B}|\psi\rangle = \mathcal{B}\mathcal{A}|\psi\rangle$ for any quantum state $|\psi\rangle$. Operators usually do not commute (rotations around different axes for example).

If they do, we can *measure* them in an arbitrary order and get the same answer. The Suzuki-Trotter expansion, however, does not require operators to commute. This has important implications for QAOA.

QAOA Hamiltonian

If $\mathcal{U}(0, T)$ has the form $\mathrm{e}^{\mathcal{A}+\mathcal{B}}$, then the Suzuki-Trotter expansion yields

$$\mathrm{e}^{\mathcal{A}+\mathcal{B}} = \lim_{m\uparrow\infty}\left(\mathrm{e}^{\frac{\mathcal{A}}{m}}\mathrm{e}^{\frac{\mathcal{B}}{m}}\right)^m \approx \mathrm{e}^{\frac{\mathcal{A}}{m}}\mathrm{e}^{\frac{\mathcal{B}}{m}}\mathrm{e}^{\frac{\mathcal{A}}{m}}\mathrm{e}^{\frac{\mathcal{B}}{m}}\cdots\mathrm{e}^{\frac{\mathcal{A}}{m}}\mathrm{e}^{\frac{\mathcal{B}}{m}}.$$

The time evolution of $\mathcal{A}+\mathcal{B}$ can be approximated by applying alternatively $\mathcal{A}$ and $\mathcal{B}$ for time intervals T/m.

The Hamiltonian terms $\mathcal{H}_0$ and $\mathcal{H}_F$ have the following general form:

$$\mathcal{H}_0 = \sum_i \sigma_x^i \qquad \text{and} \qquad \mathcal{H}_F = \sum_i a_i\sigma_z^i + \sum_{ij} b_{ij}\sigma_z^i\sigma_z^j,$$

where a_i and b_{ij} are some coefficients.

The initial Hamiltonian $\mathcal{H}_0$ is the operator $\mathcal{A}$ (the *mixing* Hamiltonian) and the final Hamiltonian $\mathcal{H}_F$ is the operator $\mathcal{B}$ (the *phase* Hamiltonian).

Quantum Approximate Optimisation Algorithm

Algorithm 1: Quantum Approximate Optimisation Algorithm

Result: Optimal solution

- Create a parameterised quantum state $|\psi(\beta,\gamma)\rangle$ by alternately applying $\mathcal{A}$ and $\mathcal{B}$ for m rounds, where the duration in round i ($i = 1, \ldots, m$) is specified by β_i and γ_i:

$$|\psi(\beta,\gamma)\rangle = \mathrm{e}^{-\mathrm{i}\beta_m \mathcal{A}} \mathrm{e}^{-\mathrm{i}\gamma_m \mathcal{B}} \cdots \mathrm{e}^{-\mathrm{i}\beta_2 \mathcal{A}} \mathrm{e}^{-\mathrm{i}\gamma_2 \mathcal{B}} \mathrm{e}^{-\mathrm{i}\beta_1 \mathcal{A}} \mathrm{e}^{-\mathrm{i}\gamma_1 \mathcal{B}} |0\rangle .$$

- A computational basis measurement is performed on $|\psi(\beta,\gamma)\rangle$, which returns a candidate solution. Repeat this state preparation and measurement and return the expected value of the cost function f over the returned solution samples:

$$\langle f\rangle = \langle \psi(\beta,\gamma)|\mathcal{B}|\psi(\beta,\gamma)\rangle ,$$

  which can be statistically estimated from the samples produced.
- The above steps may then be repeated with the updated sets of time parameters β and γ – the variational part of the algorithm – within the classical optimisation loop that aims to minimise the expectation of the cost function $\langle f\rangle$.
- The algorithm returns the best found solution.

Operators $\mathcal{A}$ and $\mathcal{B}$ must not commute

- Important to apply $\exp\{-\mathrm{i}\beta\mathcal{A}\}$ and $\exp\{-\mathrm{i}\gamma\mathcal{B}\}$ alternately to ensure that we are not trapped in a local minimum.
- Important that $\mathcal{A}$ and $\mathcal{B}$ do not commute, otherwise we may get stuck in a common eigenstate.
- For example, σ_x and σ_z do not commute:

$$\sigma_x\sigma_z - \sigma_z\sigma_x = -2\mathrm{i}\sigma_y \neq 0,$$

so that there is always a chance to escape from the local minimum.

Example : the Max-Cut problem

GOAL: divide the vertices of a graph in two groups such that either the maximum possible number of edges going between the two groups are "cut" (if all edges have the same weight) or the total weight of these edges is maximised (if they have different weights).

- Cost function for the edge connecting vertices i and j:

$$c_{ij} = \frac{1}{2} w_{ij}(1 - s_i s_j),$$

  where s_i and s_j are classical spin variables taking values $\{+1, -1\}$ and w_{ij} is the weight associated with the edge connecting vertices i and j. The two groups of vertices are those where the spin variables take the same values (either $+1$ or -1).

- Cost function for the whole graph:

$$L(\mathbf{s}) = \sum_{\{ij\} \in G} \frac{1}{2} w_{ij}(1 - s_i s_j),$$

  where $\mathbf{s} = (s_1, \ldots, s_N)$ is the set of decision variables associated with the N-node graph G and the sum goes over all pairs of nodes connected by the graph edges.

QAOA gates

The mixing Hamiltonian $\mathcal{A}$ and the phase Hamiltonian $\mathcal{B}$ that correspond to the Max-Cut cost function are

$$\mathcal{A} = \sum_{i=1}^{N} \sigma_x^i \qquad \text{and} \qquad \mathcal{B} = \sum_{\{ij\}\in G} \frac{1}{2} w_{ij} \left(1 - \sigma_z^i \sigma_z^j\right),$$

where the spin variables s are replaced by the corresponding Pauli operators σ. We thus need to find the gate representation of the operators

$$\exp\left\{-\mathrm{i}\beta\sigma_x^i\right\} \qquad \text{and} \qquad \exp\left\{-\frac{1}{2}\mathrm{i}\gamma\sigma_z^i\sigma_z^j\right\}.$$

QAOA gates (continued)

In order to find the quantum gate representation of the operators

$$\exp\left\{-\mathrm{i}\beta\sigma_x^i\right\} \qquad \text{and} \qquad \exp\left\{-\frac{1}{2}\mathrm{i}\gamma\sigma_z^i\sigma_z^j\right\},$$

we use the following:

For any operator $\mathcal{H}$ satisfying $\mathcal{H}^2 = \mathcal{I}$,

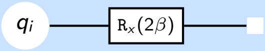
$$R_\theta(\mathcal{H}) := \exp\left\{-\mathrm{i}\frac{\theta}{2}\mathcal{H}\right\} = \cos\left(\frac{\theta}{2}\right)\mathcal{I} - \mathrm{i}\sin\left(\frac{\theta}{2}\right)\mathcal{H},$$

where $\mathcal{I}$ is the identity operator.

Proof: Taylor expansion

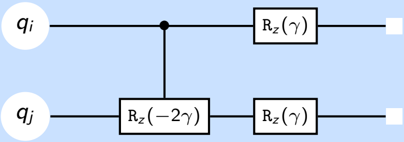
QAOA gates (continued)

$$\exp\left\{-\mathrm{i}\beta\sigma_x^i\right\} \longrightarrow$$



Gate representation of operator $\exp\left\{-\mathrm{i}\beta\sigma_x^i\right\}$.

$$\exp\left(-\mathrm{i}\frac{\gamma}{2}\sigma_z^i\sigma_z^j\right) \longrightarrow$$

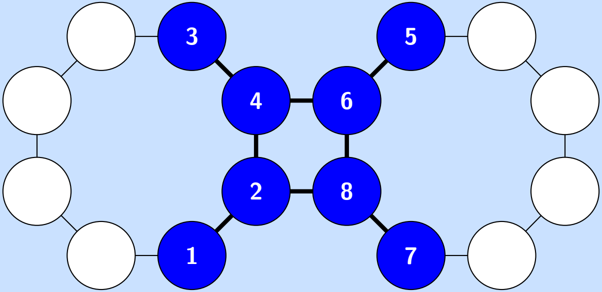


Gate representation of operator $\exp\left(-\mathrm{i}\frac{\gamma}{2}\sigma_z^i\sigma_z^j\right)$.

Sample Max-Cut problem

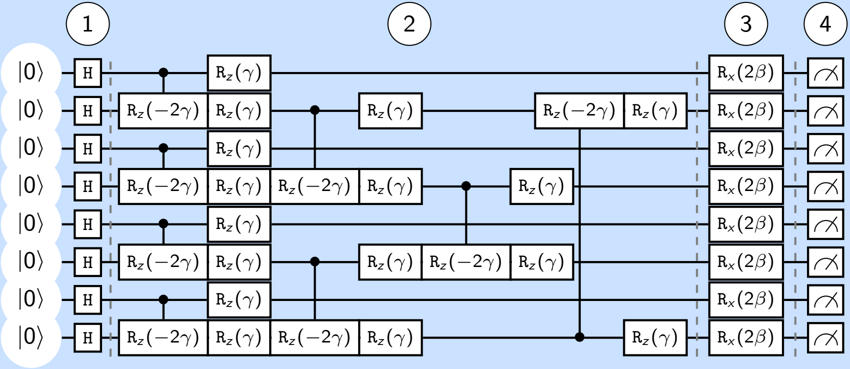
The Max-Cut graph consists of 8 nodes (qubits $1, \dots, 8$) and 8 edges with equal weights.

Problem: Find a partition that cuts the maximal number of edges.



Embedding of Max-Cut optimisation problem on Rigetti's Aspen system.

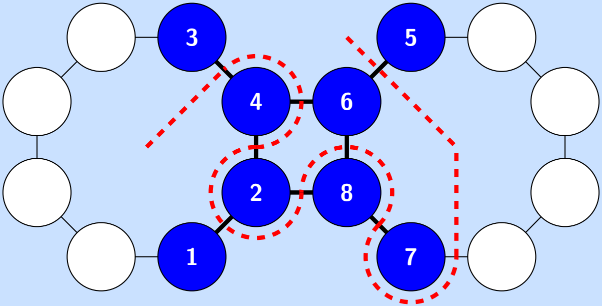
QAOA circuit



QAOA circuit for the Max-Cut problem.

Max-Cut solution

The optimal solution reads as 10011001 and is represented by the dashed curve that separates nodes into two equal subsets and cuts across all edges of the graph.



Visualisation of the Max-Cut problem solution.